

Proposal of Mach-Zehnder Interferometer

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I. INTRODUCTION

THIS is a simple proposal for a Mach-Zehnder interferometer (MZI). A Mach-Zehnder interferometer is a device that splits the input light into two paths, and then recombines them allowing us to measure phase difference between the two trajectories. The phase difference encodes information about how light traveled through both paths. Possible applications go from sensors to optical switches and high-speed modulators.

II. THEORY

Let us consider a simple model for the Mach-Zehnder interferometer. The following model applies to single mode waveguides where we disregard the field distribution and only consider the total intensity within each waveguide. The input intensity (I_i) is split into two paths by a Y-branch splitter, resulting in the input intensity (E_i) being divided in two equal parts $E_1 = E_i/\sqrt{2}$ and $E_2 = E_i/\sqrt{2}$. The two paths are characterized by their respective lengths L_1 and $L_2 = L_1 + \Delta L$, the coefficients $\beta_1 = 2\pi n_1/\lambda$ and $\beta_2 = 2\pi n_2/\lambda$ where n_1 and n_2 are the effective indices of the waveguides, and λ is the wavelength of the light and assume negligible losses. It follows that the output intensity (I_o) is given by the following equation:

$$I_o = \frac{I_i}{2} (1 + \cos(\beta_1 L_1 - \beta_2 L_2)) \quad (1)$$

For identical waveguides, we have $n_1 = n_2 = n_{eff}$ and $\beta_1 = \beta_2 = \beta$, and the free spectral range (FSR) is given by the following equation:

$$FSR = \frac{\lambda^2}{n_g \Delta L}, \quad (2)$$

where n_g is the group index of the waveguide. The FSR is the wavelength separation between two consecutive maxima in the output intensity.

III. MODELING AND SIMULATION

The proposed design is very simple. We assume a single mode TE polarized light at a wavelength of 1550 nm entering the interferometer through a Bragg grating coupler. The light is immediately split into two paths by a Y-branch splitter. The two paths are designed to be identical waveguides, 500nm width and 220nm height, except for a length difference. The longer path is 152.78 μm , while the shorter path is 73.76 μm , resulting in a length difference $\Delta L = 79.02 \mu\text{m}$. The layout of the interferometer is shown in Fig. 1. The two paths are recombined by a broad band 50/50 directional coupler. The output is collected by two Bragg grating couplers. The design

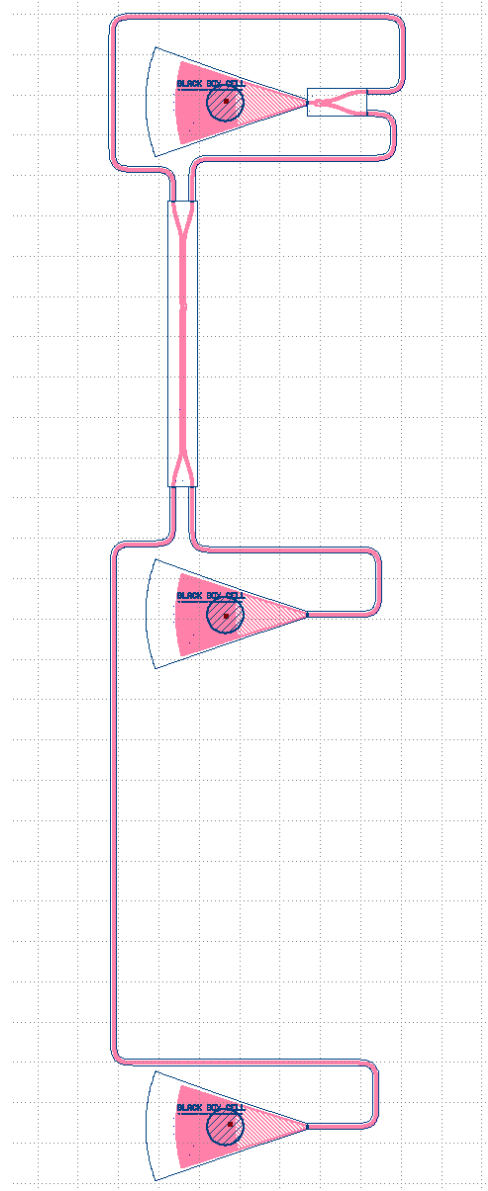


Fig. 1. Design of the Mach-Zehnder interferometer in KLayout using the SiEPIC-Ebeam library.

is implemented in KLayout and the simulation is performed using Lumerical MODE.

The mode profile of the waveguide for TE and TM polarizations are shown in Fig. 2. The effective index of the waveguide is also simulated in Lumerical MODE and the results are shown in Fig. 3. The group index was calculated

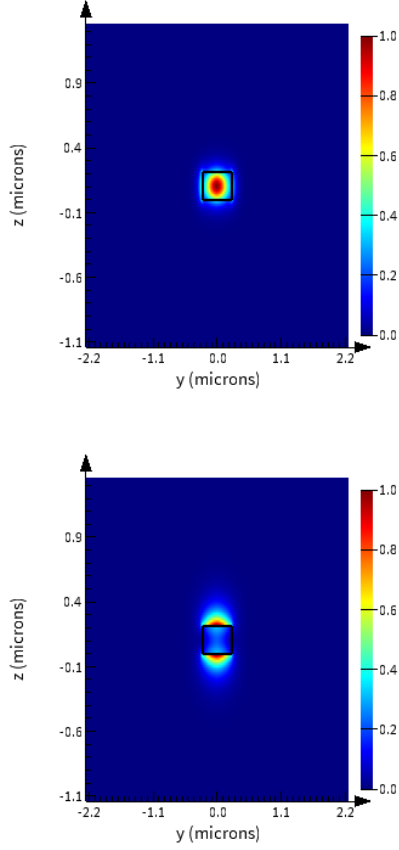


Fig. 2. Mode profiles of a 500 nm wide and 220 nm high waveguide: TE polarization (top) and TM polarization (bottom).

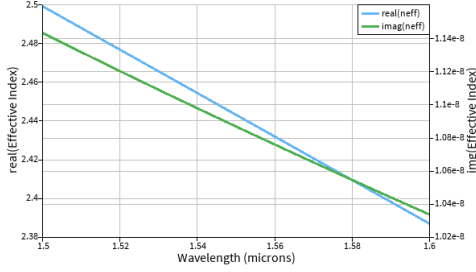


Fig. 3. Effective index of a 500 nm wide and 220 nm high waveguide as a function of wavelength.

using MATLAB, with the compact model of the effective index as a function of wavelength of the form:

$$n_{eff}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2, \quad (3)$$

with $n_1 = 2.44$, $n_2 = -1.1273$, and $n_3 = -0.03335$. The group index is shown in Fig. 4.

The simulation of the Mach-Zehnder interferometer was performed in Lumerical Interconnect. The intensity at the output ports is shown in Fig. 5. Using equation 2 and the group index calculated from the effective index, we obtain a free spectral range of 7.26 nm. Other length difference was also implemented in KLayout, see Table I.

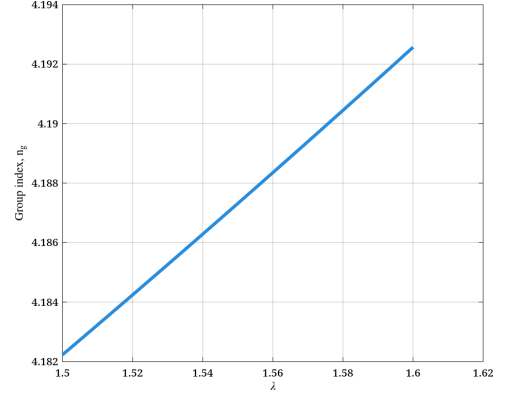


Fig. 4. Group index of a 500 nm wide and 220 nm high waveguide as a function of wavelength.

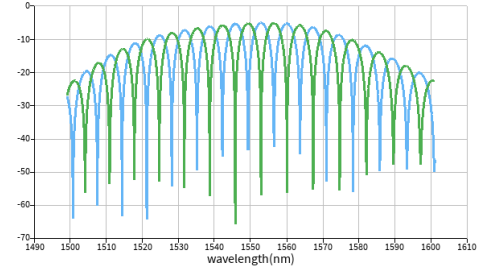


Fig. 5. Spectrum of the Mach-Zehnder interferometer.

ΔL (μm)	FSR (nm)
419.96	1.37
79.02	7.26

TABLE I

FREE SPECTRAL RANGE FOR DIFFERENT PATH LENGTH DIFFERENCES AT $\lambda = 1550$ nm.